

Solutions to Axler's *Linear Algebra Done Right*, 4th Edition

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2 Finite-Dimensional Vector Spaces

2A Span and Linear Independence

Exercise 2A1

Find a list of four distinct vectors in $\mathbb{F}^3$ whose span equals

$$\{(x, y, z) \in \mathbb{F}^3 \mid x + y + z = 0\}$$

Solution. Let $U = \{(x, y, z) \in \mathbb{F}^3 \mid x + y + z = 0\}$. Rewriting $z = -x - y$, we have

$$U = \{(x, y, -x - y) \mid x, y \in \mathbb{F}\}.$$

Consider the list of vectors

$$v_1 = (1, 0, -1), \quad v_2 = (0, 1, -1), \quad v_3 = (2, -1, -1), \quad v_4 = (1, 1, -2).$$

For any $(z, w, -z - w) \in U$, we may write it as $zv_1 + wv_2$. Therefore, v_1, v_2, v_3, v_4 span U . ■

Exercise 2A2

Prove or give a counterexample: If v_1, v_2, v_3, v_4 spans V , then the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

also spans V .

Solution. Let $U = \{v_1, v_2, v_3, v_4\}$ and $W = \{v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4\}$. Clearly, $v_4 \in \text{span } U$. Then we may write

$$\begin{aligned} v_3 &= (v_3 - v_4) + v_4 \in \text{span } W, \\ v_2 &= (v_2 - v_3) + v_3 \in \text{span } W, \\ v_1 &= (v_1 - v_2) + v_2 \in \text{span } W. \end{aligned}$$

Thus, $v_1, v_2, v_3, v_4 \in \text{span } W$, and so $V = \text{span } U \subseteq \text{span } W$. Since W is a list of vectors in V , we have $\text{span } W \subseteq V$. Therefore, $\text{span } W = V$, and so W spans V . ■

Exercise 2A3

Suppose $v_1, \dots, v_m$ is a list of vectors in V . For $k \in \{1, \dots, m\}$, let

$$w_k = v_1 + \dots + v_k.$$

Show that $\text{span}(v_1, \dots, v_m) = \text{span}(w_1, \dots, w_m)$.

Solution. This is a generalization of [Exercise 2A2](#) and a direct consequence of the fact that $v_k = w_k - w_{k-1}$ for $k \in \{2, \dots, m\}$ and $v_1 = w_1$. Since each v_k is a linear combination of $w_1, \dots, w_k$, we have

$\text{span}(v_1, \dots, v_m) \subseteq \text{span}(w_1, \dots, w_m)$. On the other hand, each w_k is a linear combination of $v_1, \dots, v_k$, so $\text{span}(w_1, \dots, w_m) \subseteq \text{span}(v_1, \dots, v_m)$. Therefore, $\text{span}(v_1, \dots, v_m) = \text{span}(w_1, \dots, w_m)$. ■

Exercise 2A4

- (a) Show that a list of length one in a vector space is linearly independent if and only if the vector in the list is not 0.
- (b) Show that a list of length two in a vector space is linearly independent if and only if neither of the two vectors in the list is a scalar multiple of the other.

Solution.

- (a) Let v be the vector in the list. If $v = 0$, then the list is linearly dependent, since $0 \cdot v = 0$. If $v \neq 0$, then the only way to write 0 as a linear combination of v is to take the scalar multiple to be 0. Thus, the list is linearly independent.
- (b) Let v_1 and v_2 be the vectors in the list. If v_1 is a scalar multiple of v_2 , then there exists a scalar λ such that $v_1 = \lambda v_2$. Then we may write

$$0 = v_1 - \lambda v_2,$$

which shows that the list is linearly dependent. Conversely, if the list is linearly dependent, then there exist scalars α and β , not both zero, such that

$$\alpha v_1 + \beta v_2 = 0.$$

If $\alpha \neq 0$, then

$$v_1 = -\frac{\beta}{\alpha} v_2.$$

Similarly, if $\beta \neq 0$, then

$$v_2 = -\frac{\alpha}{\beta} v_1.$$

Thus, the list is linearly independent if and only if neither vector is a scalar multiple of the other. ■

Exercise 2A5

Find a number t such that

$$(3, 1, 4), (2, -3, 5), (5, 9, t)$$

is not linearly independent in $\mathbb{R}^3$.

Solution. To say that the list is not linearly independent is to say that there exist scalars α, β, γ , not all zero, such that

$$\alpha(3, 1, 4) + \beta(2, -3, 5) + \gamma(5, 9, t) = (0, 0, 0).$$

This is equivalent to the following system of equations:

$$0 = 3\alpha + 2\beta + 5\gamma,$$

$$0 = \alpha - 3\beta + 9\gamma,$$

$$0 = 4\alpha + 5\beta + t\gamma.$$

From the second equation, we obtain $\alpha = 3\beta - 9\gamma$. Substituting this into the first equation, we have

$$\begin{aligned} 0 &= 3(3\beta - 9\gamma) + 2\beta + 5\gamma \\ &= 11\beta - 22\gamma. \end{aligned}$$

Thus, $\beta = 2\gamma$. Substituting this into the expression for α , we have $\alpha = 3(2\gamma) - 9\gamma = -3\gamma$. Substituting these values into the third equation, we have

$$\begin{aligned} 0 &= 4(-3\gamma) + 5(2\gamma) + t\gamma \\ &= -12\gamma + 10\gamma + t\gamma \\ &= (t - 2)\gamma. \end{aligned}$$

Since $\gamma \neq 0$, we must have $t - 2 = 0$, or $t = 2$. Then the list is not linearly independent when $t = 2$. ■

Exercise 2A6

Show that the list $(2, 3, 1), (1, -1, 2), (7, 3, c)$ is linearly dependent in $\mathbb{F}^3$ if and only if $c = 8$.

Solution. We may proceed similarly to [Exercise 2A5](#) to obtain the following system of equations:

$$\begin{aligned} 0 &= 2\alpha + \beta + 7\gamma, \\ 0 &= 3\alpha - \beta + 3\gamma, \\ 0 &= \alpha + 2\beta + c\gamma. \end{aligned}$$

The second equation gives us $\beta = 3\alpha + 3\gamma$. Substituting this into the first equation, we have

$$\begin{aligned} 0 &= 2\alpha + (3\alpha + 3\gamma) + 7\gamma \\ &= 5\alpha + 10\gamma. \end{aligned}$$

Thus, $\alpha = -2\gamma$. Substituting this into the expression for β , we have $\beta = 3(-2\gamma) + 3\gamma = -3\gamma$. Substituting these values into the third equation, we have

$$\begin{aligned} 0 &= (-2\gamma) + 2(-3\gamma) + c\gamma \\ &= -2\gamma - 6\gamma + c\gamma \\ &= (c - 8)\gamma. \end{aligned}$$

Since $\gamma \neq 0$, we must have $c - 8 = 0$, or $c = 8$. Then the list is linearly dependent when $c = 8$.

Conversely, if $c \neq 8$, then $(c - 8)\gamma = 0$ implies $\gamma = 0$. Then $\alpha = -2\gamma = 0$ and $\beta = -3\gamma = 0$, so the only solution to the system of equations is $\alpha = \beta = \gamma = 0$. Thus, the list is linearly independent when $c \neq 8$. ■

Exercise 2A7

- Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{R}$, then the list $1 + i, 1 - i$ is linearly independent.
- Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{C}$, then the list $1 + i, 1 - i$ is linearly dependent.

Solution.

(a) Suppose $\alpha, \beta \in \mathbb{R}$ such that

$$\alpha(1 + i) + \beta(1 - i) = 0.$$

Then we have

$$(\alpha + \beta) + (\alpha - \beta)i = 0.$$

Since $0 = 0 + 0i$, we must have $\alpha + \beta = 0$ and $\alpha - \beta = 0$. Solving this system of equations, we find that $\alpha = \beta = 0$. Thus, the list is linearly independent.

(b) Suppose $a + bi, c + di \in \mathbb{C}$ such that

$$(a + bi)(1 + i) + (c + di)(1 - i) = 0.$$

Then we have

$$(a - b + c + d) + (a + b - c + d)i = 0.$$

Then we have the system of equations

$$\begin{aligned} 0 &= a - b + c + d, \\ 0 &= a + b - c + d. \end{aligned}$$

Adding the two equations results in $0 = 2a + 2d$, or $a = -d$. Substituting this into the first equation, we obtain $b = c$. Then we have infinitely many nontrivial solutions to the system of equations, such as $a = 1, b = 0, c = 0, d = -1$. Thus, the list is linearly dependent. ■

Exercise 2A8

Suppose v_1, v_2, v_3, v_4 is linearly independent in V . Prove that the list $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$ is also linearly independent.

Solution. Let $\alpha, \beta, \gamma, \delta \in \mathbb{F}$ such that

$$\alpha(v_1 - v_2) + \beta(v_2 - v_3) + \gamma(v_3 - v_4) + \delta v_4 = 0.$$

Then we have

$$\alpha v_1 + (-\alpha + \beta)v_2 + (-\beta + \gamma)v_3 + (-\gamma + \delta)v_4 = 0.$$

Since v_1, v_2, v_3, v_4 is linearly independent, we must have

$$\begin{aligned} 0 &= \alpha, \\ 0 &= -\alpha + \beta, \\ 0 &= -\beta + \gamma, \\ 0 &= -\gamma + \delta. \end{aligned}$$

From the first equation, we have $\alpha = 0$. Substituting this into the second equation gives us $\beta = 0$. Substituting this into the third equation gives us $\gamma = 0$. Finally, substituting this into the fourth equation gives us $\delta = 0$. Thus, the only solution to the original equation is $\alpha = \beta = \gamma = \delta = 0$, and so the list is linearly independent. ■

Exercise 2A9

Prove or give a counterexample: If $v_1, v_2, \dots, v_m$ is a linearly independent list of vectors in V , then $5v_1 - 4v_2, v_2, v_3, \dots, v_m$ is linearly independent.

Solution. Let $\alpha_1, \dots, \alpha_m \in \mathbb{F}$ such that

$$\alpha_1(5v_1 - 4v_2) + \alpha_2v_2 + \dots + \alpha_mv_m = 0.$$

Then we have

$$5\alpha_1v_1 + (-4\alpha_1 + \alpha_2)v_2 + \alpha_3v_3 + \dots + \alpha_mv_m = 0.$$

Since $v_1, v_2, \dots, v_m$ is linearly independent, we must have

$$\begin{aligned} 0 &= 5\alpha_1, \\ 0 &= -4\alpha_1 + \alpha_2, \\ 0 &= \alpha_3, \\ &\vdots \\ 0 &= \alpha_m. \end{aligned}$$

From the first equation, we have $\alpha_1 = 0$. Substituting this into the second equation gives us $\alpha_2 = 0$. The remaining equations give us $\alpha_k = 0$ for $k = 3, \dots, m$. Thus, the only solution to the original equation is $\alpha_k = 0$ for $k = 1, \dots, m$, and so the list is linearly independent. ■

Exercise 2A10

Prove or give a counterexample: If $v_1, v_2, \dots, v_m$ is a linearly independent list of vectors in V and $\lambda \in \mathbb{F}$ with $\lambda \neq 0$, then $\lambda v_1, \lambda v_2, \dots, \lambda v_m$ is linearly independent.

Solution. Let $\alpha_1, \dots, \alpha_m \in \mathbb{F}$ such that

$$\alpha_1(\lambda v_1) + \dots + \alpha_m(\lambda v_m) = 0.$$

Then we have

$$\lambda(\alpha_1v_1 + \dots + \alpha_mv_m) = 0.$$

Since $\lambda \neq 0$, we must have

$$\alpha_1v_1 + \dots + \alpha_mv_m = 0.$$

Since $v_1, v_2, \dots, v_m$ is linearly independent, we must have $\alpha_k = 0$ for $k = 1, \dots, m$. Thus, the only solution to the original equation is $\alpha_k = 0$ for $k = 1, \dots, m$, and so the list is linearly independent. ■

Exercise 2A11

Prove or give a counterexample: If $v_1, \dots, v_m$ and $w_1, \dots, w_m$ are linearly independent lists of vectors in V , then the list $v_1 + w_1, \dots, v_m + w_m$ is linearly independent.

Solution. Consider the following counterexample in $\mathbb{R}^2$:

$$v_1 = (1, 0), \quad v_2 = (0, 1), \quad w_1 = (0, 1), \quad w_2 = (1, 0).$$

Then v_1, v_2 and w_1, w_2 are linearly independent lists of vectors in $\mathbb{R}^2$. However,

$$v_1 + w_1 = (1, 1), \quad v_2 + w_2 = (1, 1),$$

and so the list $v_1 + w_1, v_2 + w_2$ is linearly dependent. Thus, the statement is false. ■

Exercise 2A12

Suppose $v_1, \dots, v_m$ is linearly independent in V and $w \in V$. Prove that if $v_1 + w, \dots, v_m + w$ is linearly dependent, then $w \in \text{span}(v_1, \dots, v_m)$.

Solution. Let $\alpha_1, \dots, \alpha_m \in \mathbb{F}$, not all zero, such that

$$\alpha_1(v_1 + w) + \dots + \alpha_m(v_m + w) = 0.$$

Then we have

$$(\alpha_1 v_1 + \dots + \alpha_m v_m) + (\alpha_1 + \dots + \alpha_m)w = 0.$$

If $\alpha_1 + \dots + \alpha_m = 0$, then we have

$$\alpha_1 v_1 + \dots + \alpha_m v_m = 0,$$

which contradicts the linear independence of $v_1, \dots, v_m$. Thus, we must have $\alpha_1 + \dots + \alpha_m \neq 0$. Let $\beta = \alpha_1 + \dots + \alpha_m$. Then we may write

$$w = -\frac{\alpha_1}{\beta}v_1 - \dots - \frac{\alpha_m}{\beta}v_m.$$

Thus, w is a linear combination of $v_1, \dots, v_m$, and so $w \in \text{span}(v_1, \dots, v_m)$. ■

Exercise 2A13

Suppose $v_1, \dots, v_m$ is linearly independent in V and $w \in V$. Show that

$$v_1, \dots, v_m, w \text{ is linearly independent} \iff w \notin \text{span}(v_1, \dots, v_m).$$

Solution. Suppose $v_1, \dots, v_m, w$ is linearly independent. If there were scalars $\alpha_1, \dots, \alpha_m, \beta$ such that

$$\alpha_1 v_1 + \dots + \alpha_m v_m + \beta w = 0,$$

then linear independence would imply that $\alpha_1 = \dots = \alpha_m = \beta = 0$. Suppose $w \in \text{span}(v_1, \dots, v_m)$. Then there exist scalars $\gamma_1, \dots, \gamma_m$ such that

$$\gamma_1 v_1 + \dots + \gamma_m v_m = w.$$

Then we may write

$$-\gamma_1 v_1 - \dots - \gamma_m v_m + 1 \cdot w = 0,$$

which contradicts the linear independence of $v_1, \dots, v_m, w$. Thus, $w \notin \text{span}(v_1, \dots, v_m)$.

Conversely, suppose $w \notin \text{span}(v_1, \dots, v_m)$. Then there do not exist scalars $\alpha_1, \dots, \alpha_m$ such that

$$\alpha_1 v_1 + \dots + \alpha_m v_m = w.$$

If $v_1, \dots, v_m, w$ is linearly dependent, then there exist scalars $\beta_1, \dots, \beta_m, \gamma$, not all zero, such that

$$\beta_1 v_1 + \dots + \beta_m v_m + \gamma w = 0.$$

If $\gamma = 0$, then we have

$$\beta_1 v_1 + \dots + \beta_m v_m = 0,$$

which contradicts the linear independence of $v_1, \dots, v_m$. Thus, we must have $\gamma \neq 0$. Then we may write

$$w = -\frac{\beta_1}{\gamma} v_1 - \dots - \frac{\beta_m}{\gamma} v_m,$$

which contradicts the assumption that w is not in the span of $v_1, \dots, v_m$. Thus, $v_1, \dots, v_m, w$ is linearly independent. ■

Exercise 2A14

Suppose $v_1, \dots, v_m$ is a list of vectors in V . For $k \in \{1, \dots, m\}$, let

$$w_k = v_1 + \dots + v_k.$$

Show that the list $v_1, \dots, v_m$ is linearly independent if and only if the list $w_1, \dots, w_m$ is linearly independent.

Solution. Suppose $v_1, \dots, v_m$ is linearly independent. Let $\alpha_1, \dots, \alpha_m \in \mathbb{F}$ such that

$$\alpha_1 w_1 + \dots + \alpha_m w_m = 0.$$

Substituting $w_k = v_1 + \dots + v_k$ for each k , we get

$$\alpha_1 v_1 + \alpha_2(v_1 + v_2) + \alpha_3(v_1 + v_2 + v_3) + \dots + \alpha_m(v_1 + \dots + v_m) = 0.$$

Then we may rewrite this as

$$(\alpha_1 + \dots + \alpha_m)v_1 + (\alpha_2 + \dots + \alpha_m)v_2 + \dots + (\alpha_{m-1} + \alpha_m)v_{m-1} + \alpha_m v_m = 0.$$

Since $v_1, \dots, v_m$ is linearly independent, we must have $\alpha_1 = \dots = \alpha_m = 0$. Hence, $w_1, \dots, w_m$ is linearly independent.

Conversely, suppose $w_1, \dots, w_m$ is linearly independent. Let $\beta_1, \dots, \beta_m \in \mathbb{F}$ such that

$$\beta_1 v_1 + \dots + \beta_m v_m = 0.$$

Noting that $v_1 = w_1$ and $v_k = w_k - w_{k-1}$ for $k \in \{2, \dots, m\}$, we may rewrite the equation as

$$\begin{aligned} \beta_1 w_1 + \beta_2(w_2 - w_1) + \dots + \beta_m(w_m - w_{m-1}) &= 0, \\ (\beta_1 - \beta_2)w_1 + (\beta_2 - \beta_3)w_2 + \dots + (\beta_{m-1} - \beta_m)w_{m-1} + \beta_m w_m &= 0. \end{aligned}$$

Since $w_1, \dots, w_m$ is linearly independent, we find that the system of equations

$$\begin{aligned} 0 &= \beta_1 - \beta_2, \\ 0 &= \beta_2 - \beta_3, \\ &\vdots \\ 0 &= \beta_{m-1} - \beta_m \\ 0 &= \beta_m, \end{aligned}$$

has the unique solution $\beta_1 = \dots = \beta_m = 0$. Hence, $v_1, \dots, v_m$ is linearly independent. ■

Exercise 2A15

Explain why there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(\mathbb{F})$.

Solution. Since $\mathcal{P}_4(\mathbb{F})$ can be spanned by the list $1, x, x^2, x^3, x^4$, and the length of a linearly independent list cannot exceed the length of a spanning list, there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(\mathbb{F})$. ■

Exercise 2A16

Explain why no list of four polynomials spans $\mathcal{P}_4(\mathbb{F})$.

Solution. Since the list $1, x, x^2, x^3, x^4$ is a linearly independent list in $\mathcal{P}_4(\mathbb{F})$ and has length 5, and the length of a spanning list cannot be less than the length of a linearly independent list, no list of four polynomials can span $\mathcal{P}_4(\mathbb{F})$. ■

Exercise 2A17

Prove that V is infinite-dimensional if and only if there is a sequence $v_1, v_2, \dots$ of vectors in V such that $v_1, \dots, v_m$ is linearly independent for every positive integer m .

Solution. Suppose V is infinite-dimensional. We proceed by induction to construct a sequence of vectors $v_1, v_2, \dots$ in V such that $v_1, \dots, v_m$ is linearly independent for every positive integer m . Since V is infinite-dimensional, there exists a nonzero vector $v_1 \in V$. Suppose we have constructed a linearly independent list of vectors $v_1, \dots, v_m$ in V . If every $v \in V$ is a linear combination of $v_1, \dots, v_m$, then $V = \text{span}(v_1, \dots, v_m)$, which contradicts the assumption that V is infinite-dimensional. Thus, there exists a vector $v_{m+1} \in V$ that is not in $\text{span}(v_1, \dots, v_m)$. Then by [Exercise 2A13](#), the list $v_1, \dots, v_m, v_{m+1}$ is linearly independent. By induction, we can construct a sequence of vectors $v_1, v_2, \dots$ in V such that $v_1, \dots, v_m$ is linearly independent for every positive integer m .

Conversely, suppose there exists a sequence of vectors $v_1, v_2, \dots$ in V such that $v_1, \dots, v_m$ is linearly independent for every positive integer m . Then for every positive integer m , there exists a linearly independent list of vectors in V of length m . Since there is no maximum length for a linearly independent list of vectors in V , this implies that V is infinite-dimensional. ■

Exercise 2A18

Prove that $\mathbb{F}^\infty$ is infinite-dimensional.

Solution. By [Exercise 2A17](#), it suffices to construct a sequence of vectors $v_1, v_2, \dots$ in $\mathbb{F}^\infty$ such that $v_1, \dots, v_m$ is linearly independent for every positive integer m . Let v_k be the vector in $\mathbb{F}^\infty$ whose k -th coordinate is 1 and all other coordinates are 0. Explicitly, we have

$$v_1 = (1, 0, 0, \dots), \quad v_2 = (0, 1, 0, \dots), \quad v_3 = (0, 0, 1, \dots), \quad \dots$$

Then for any positive integer m , the list $v_1, \dots, v_m$ is linearly independent, since the only way to write the zero vector as a linear combination of $v_1, \dots, v_m$ is to take all coefficients to be zero. Thus, by [Exercise 2A17](#), $\mathbb{F}^\infty$ is infinite-dimensional. ■

Exercise 2A19

Prove that the real vector space of all continuous real-valued functions on the interval $[0, 1]$ is infinite-dimensional.

Solution. By [Exercise 2A17](#), it suffices to construct a sequence of continuous real-valued functions $f_1, f_2, \dots$ on the interval $[0, 1]$ such that $f_1, \dots, f_m$ is linearly independent for every positive integer m . Let $f_k(x) = x^{k-1}$ for each positive integer k . Then for any positive integer m , let $\alpha_1, \dots, \alpha_m \in \mathbb{R}$ such that

$$\alpha_1 f_1(x) + \dots + \alpha_m f_m(x) = 0$$

for all $x \in [0, 1]$. This is equivalent to the polynomial equation

$$\alpha_1 + \alpha_2 x + \dots + \alpha_m x^{m-1} = 0$$

for all $x \in [0, 1]$. Since a polynomial of degree at most $m-1$ can have at most $m-1$ roots unless it is the zero polynomial, we must have $\alpha_1 = \dots = \alpha_m = 0$. Thus, the list $f_1, \dots, f_m$ is linearly independent. Therefore, by [Exercise 2A17](#), the real vector space of all continuous real-valued functions on the interval $[0, 1]$ is infinite-dimensional. ■

Exercise 2A20

Suppose $p_0, p_1, \dots, p_m$ are polynomials in $\mathcal{P}_m(\mathbb{F})$ such that $p_k(2) = 0$ for each $k \in \{0, \dots, m\}$. Prove that $p_0, p_1, \dots, p_m$ is not linearly independent in $\mathcal{P}_m(\mathbb{F})$.

Solution. Since $p_k(2) = 0$ for each $k \in \{0, \dots, m\}$, we have

$$p_k(x) = (x - 2)q_k(x)$$

for some polynomial $q_k(x) \in \mathcal{P}_{m-1}(\mathbb{F})$. Then we may write

$$p_0(x), p_1(x), \dots, p_m(x) = (x - 2)q_0(x), (x - 2)q_1(x), \dots, (x - 2)q_m(x).$$

Since $\mathcal{P}_{m-1}(\mathbb{F})$ can be spanned by the m polynomials $1, x, \dots, x^{m-1}$, the list of $m+1$ polynomials $q_0, q_1, \dots, q_m$ must be linearly dependent. Thus, there exist scalars $\alpha_0, \dots, \alpha_m$, not all zero, such that

$$\alpha_0 q_0(x) + \dots + \alpha_m q_m(x) = 0.$$

Then we may write

$$\alpha_0 p_0(x) + \cdots + \alpha_m p_m(x) = (x - 2)(\alpha_0 q_0(x) + \cdots + \alpha_m q_m(x)) = 0.$$

Since not all the scalars $\alpha_0, \dots, \alpha_m$ are zero, this shows that $p_0, p_1, \dots, p_m$ is linearly dependent in $\mathcal{P}_m(\mathbb{F})$. ■